Control of Multi-joint Manipulator for Grasping on Water Surface

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Floating-base Manipulator

1. Mounting multi-joint manipulators on Unmanned Surface Vehicles (USVs) empowers USVs to perform complex operations.
2. Control of the floating-base manipulator is challenging due to the undesired base disturbance

Figure: Floating-base manipulator mounted on aquatic platform.

System Dynamics

Kosuge et al.(1998): Control by deriving system kinematics and dynamics. Hard for high degree-of-freedom system due to complexity.

Predictive Control

From et al. (2010), Xu et al. (2021): Forecast future disturbance with predictive control scheme. It avoids the cost for system modeling.

Kinematics Equivalence

Woolfrey et al. (2021): Covert floating-base problem to fixed-base problem.

Introduction

Literature Review

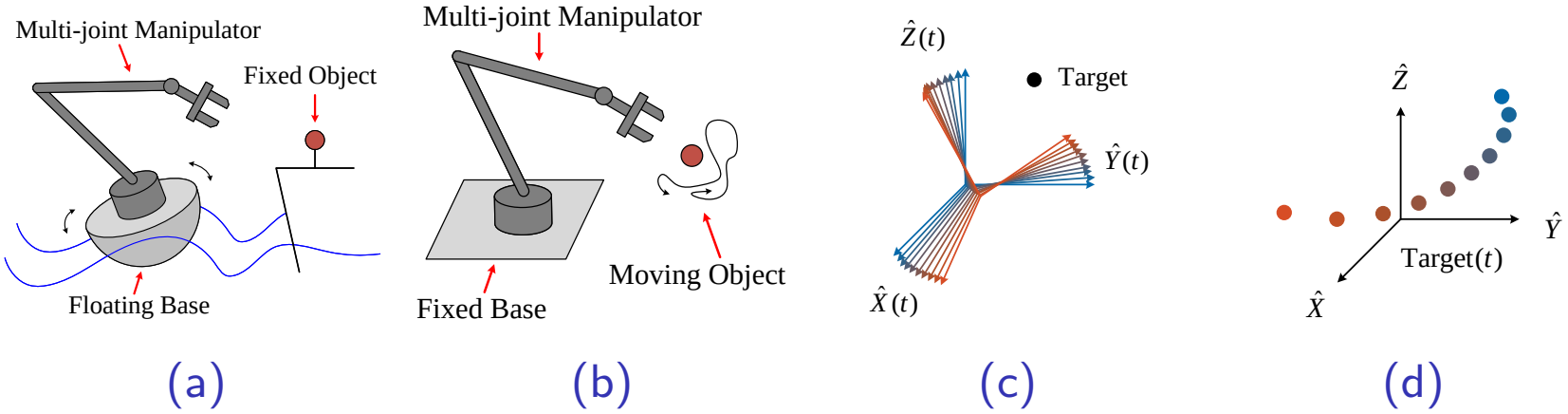


Figure: Illustration of the kinematic equivalence of (a) grasping a fixed object on floating base and (b) grasping a moving object on fixed base, as (c) a fixed target relative to a moving frame appears equivalent to (d) moving target relative to a fixed frame.

Introduction

Contribution

Motivated by:
Predictive Control + Kinematics Equivalence

This project develops a novel framework to control a multi-joint manipulator for grasping target on disturbed water surface.

Methodology

Overview

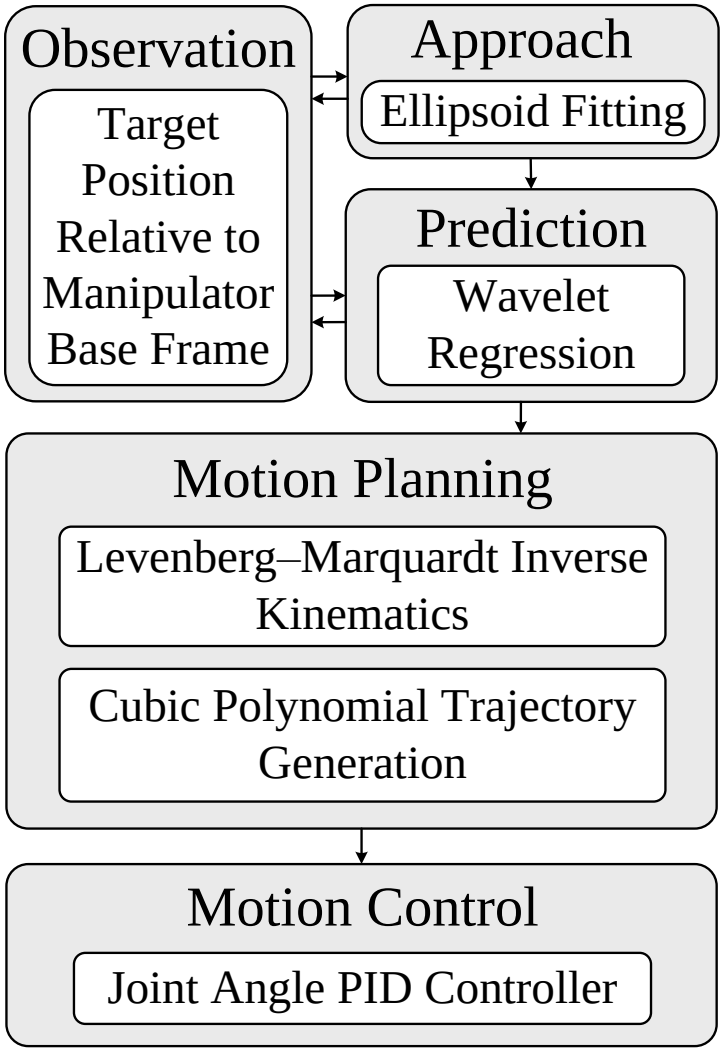


Figure: Overview of the proposed control framework.

Methodology

Approach Policy

1. Fitted a minimal volume enclosing ellipsoid (MVEE) for sampled target position.
2. Get a safety boundary by expanding MVEE.
3. Choose a proper point at safety boundary as approach position.

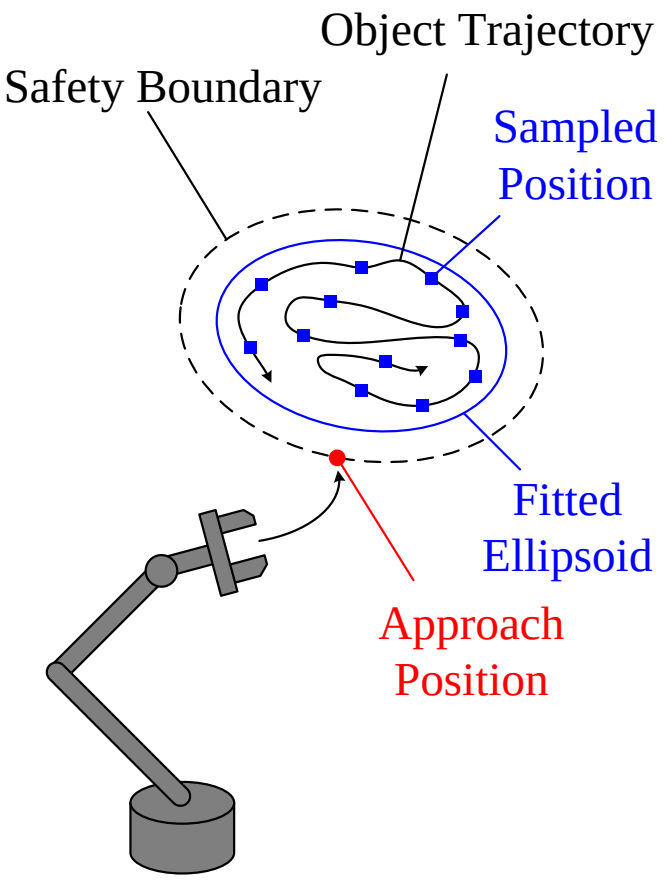


Figure: Approach policy

Methodology

Approach Policy

Ellipsoid Function:

$$(\mathbf{p}-\mathbf{c})^{\top}\mathbf{A}(\mathbf{p}-\mathbf{c})=1 \tag{1}$$

where $\mathbf{A} \in \mathbf{R}_{3\times 3}$ id the ellipsoid's determinant and $\mathbf{c} \in \mathbf{R}^3$ is the ellipsoid's center.
Sampled data: $\{\hat{\mathbf{p}}_i, i=1,2,3,\ldots,n\}$

MVEE Fitting

$$\begin{aligned} \min_{\mathbf{c},\mathbf{A}} \quad & -\log(|\mathbf{A}|) \\ s.t. \quad & (\hat{\mathbf{p}}_i-\mathbf{c})^{\top}\mathbf{A}(\hat{\mathbf{p}}_i-\mathbf{c}) \leq 1, \quad i=1,2,3,\ldots,n \end{aligned} \tag{2}$$

(2) can be solved with Khachiyan's algorithm.

Methodology

Approach Policy

Safety Boundary Generation

$$(\mathbf{p}-\mathbf{c})^{\top}\mathbf{A}(\mathbf{p}-\mathbf{c})=\lambda, \quad (\lambda>1) \tag{3}$$

Approach Position Selection

$$\begin{aligned} \min_{\mathbf{p}} \quad & ||\mathbf{p}|| \\ s.t. \quad & (\mathbf{p}-\mathbf{c})^{\top}\mathbf{A}(\mathbf{p}-\mathbf{c})=\lambda, \quad (\lambda>1) \end{aligned} \tag{4}$$

Methodology

Motion Prediction

Target position $\mathbf{p}_t = [x_t, y_t, z_t]^\top$ modeled with time series theory.

Variance reduction with ANOVA Expansion

$$\begin{aligned} x_t &= F(x_{t-1}, x_{t-2}, \ldots, x_{t-N}) \\ &= \sum_{i=t-N}^{t-1} f_i(x_i) + \sum_{i,j=t-N, i\neq j}^{t-1} f_{ij}(x_i, x_j) + e_t \end{aligned} \tag{5}$$

Wavelet Regression

$$\begin{aligned} f_i(x_i) &= \sum_k \sum_l c_{k,l,i}\ \psi^{[1]}_{k,l}(x_i) \\ f_{ij}(x_i, x_j) &= \sum_k \sum_{l_1} \sum_{l_2} c_{k,l_1,l_2,ij}\ \psi^{[2]}_{k,l_1,l_2}(x_i, x_j) \end{aligned} \tag{6}$$

Wavelet Coefficients

$$x_t = \boldsymbol{\theta}_t^\top \boldsymbol{\psi}_t(\mathbf{x}_{t,N}) + e_t \approx \boldsymbol{\theta}_t^\top \boldsymbol{\psi}_t(\mathbf{x}_{t,N}) \tag{7}$$

$\boldsymbol{\theta}_t$ is collection of wavelet coefficients and $\boldsymbol{\psi}_t$ is collection of candidate wavelet functions. Given N historical measurements $\hat{\mathbf{x}}_{t,N} = [\hat{x}_{t-N}, \hat{x}_{t-N+1}, \ldots, \hat{x}_{t-1}]^\top \in \mathbb{R}^N$, $\boldsymbol{\Psi}_{t,N} = [\boldsymbol{\psi}_{t-N}, \boldsymbol{\psi}_{t-N+1}, \ldots, \boldsymbol{\psi}_{t-1}]^\top \in \mathbb{R}^{N \times M}$, a linear system can be formulated based on (7).

$$\boldsymbol{\Psi}_{t,N}\ \boldsymbol{\theta}_t = \hat{\mathbf{x}}_{t,N} \tag{8}$$

Significant Terms Selection

For computational efficiency, (8) is solved with orthogonal least squares (OLS) algorithm. $\mathbf{\Psi}_{t,N} = \mathbf{QR}$, $\mathbf{Q} \in \mathbb{R}^{N\times N}$ is an orthogonal matrix and $\mathbf{R} \in \mathbb{R}^{N\times M}$ is an upper-triangle matrix. A subset $\mathbf{Q}_s$ of $\mathbf{Q}$ is selected from those column vectors $\mathbf{Q}_i$ of $\mathbf{Q}$ having significant contributions.

$$\mathrm{ERR}(\mathbf{Q}_i) = \hat{\mathbf{x}}_{t,N}^{\top}\mathbf{Q}_i \left(\hat{x}_{t,N}^{\top}\hat{x}_{t,N}\right)^{\top} \tag{9}$$

$$\mathbf{Q}_i = \arg\max_{\mathbf{Q}_i} \ \mathrm{ERR}(\mathbf{Q}_i), i = 1, 2, \ldots, N \tag{10}$$

Once $1 - \sum_i \mathrm{ERR}(\mathbf{Q}_i) < \epsilon, \mathbf{Q}_i \in \mathbf{Q}_s$, where ϵ is the tolerance, the selection procedure of $\mathbf{Q}_i$ terminates.

$$\mathbf{\Psi}_{t,N,s}\ \boldsymbol{\theta}_{t,s} = \hat{\mathbf{x}}_{t,N} \tag{11}$$

Single-step Prediction

$$x_t^* = \boldsymbol{\theta}_{t,s}^\top \boldsymbol{\psi}_{t,s}(\hat{\mathbf{x}}_{t,N}) \tag{12}$$

Multi-step Prediction

(12) is iterated by substituting predicted value to the historical data set and iteratively repeating process (8) to (12).

Inverse Kinematics

$$\begin{bmatrix} \mathbf{R}_T & \mathbf{p}_T \\ 0_{1\times 3} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}(\mathbf{q}_T) & \mathbf{p}(\mathbf{q}_T) \\ 0_{1\times 3} & 1 \end{bmatrix} \tag{13}$$

where $\mathbf{R}(\cdot)$ and $\mathbf{Q}(\cdot)$ are forward kinematics usually provided by manipulator's manufacturer; $\mathbf{q}_T$ is the corresponding n-DOF joint variable vector to achieve desired pose $(\mathbf{R}_T, \mathbf{p}_T)$.

Levenberg-Marquard Algorithm (LMA)

$$\mathbf{A}_{12\times n}\ \mathbf{q}_{T,1\times n} = \mathbf{b}_{1\times 12} \tag{14}$$

$$\mathbf{q}_{i+1} = \mathbf{q}_i + \lambda^{-1} J(\mathbf{q}_i)^\top (\mathbf{b} - \mathbf{A}\mathbf{q}_i) \tag{15}$$

where λ is the LMA damping factor and $J(\mathbf{q})$ is the Jacobian of $\mathbf{A}\mathbf{q}$. Iteration (15) terminates when desired precision is achieved.

Cubic Polynomial Trajectory Generation

$$
\begin{aligned}
& q_i(t) = a_{i0} + a_{i1}t + a_{i2}t^2 + a_{i3}t^3 \\
& \begin{cases} q_i(t_c) = q_{Ci} \\ \dot{q}_i(t_c) = 0 \end{cases} \quad \begin{cases} q_i(t_c + \triangle t) = q_{Ti} \\ \dot{q}_i(t_c + \triangle t) = 0 \end{cases} \\
& \quad i = 1, 2, 3, \ldots, n
\end{aligned} \tag{16}
$$

Joint velocity and acceleration constraints are assumed to be satisfied.

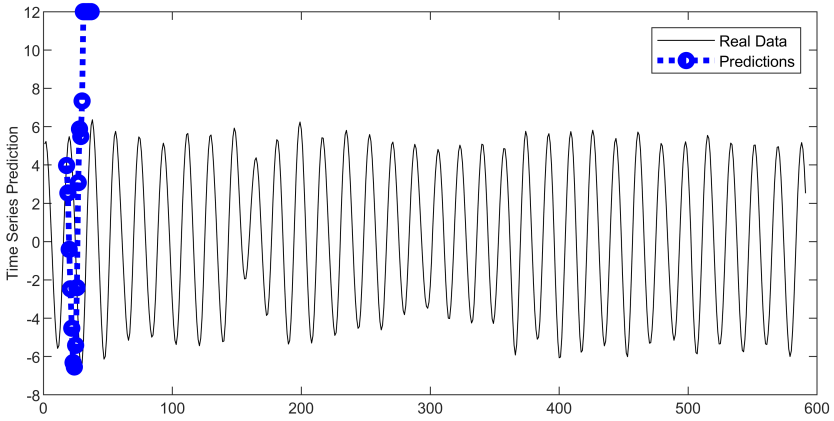
Experimental Results

Three Experiments

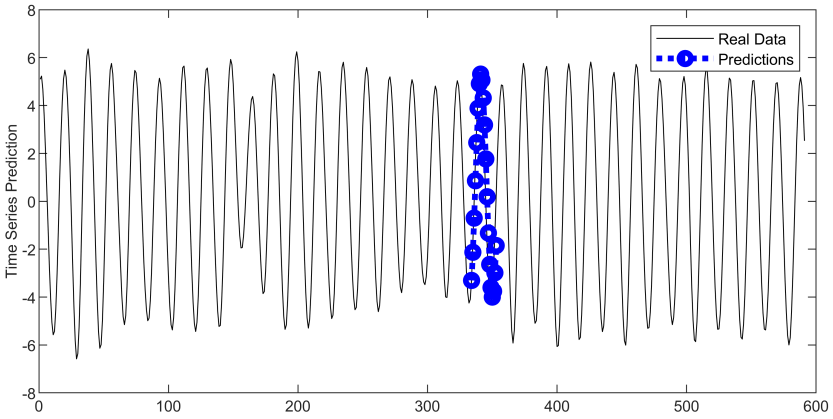
1. Experiment 1: Wavelet Prediction
2. Experiment 2: Inverse Kinematics: LMA v.s. KDL
3. Experiment 3: Grasping Demonstration
   - 3.1 MATLAB/Simulink Simscape Simulation
   - 3.2 Physical Demonstration

Experimental Results

Wavelet Prediction



(a)



(b)

Figure: Demonstration of a sliding-window multi-step prediction (sample horizon 50, prediction horizon 20) of a time series with wavelet regression. (a) In the initial stage, due to insufficient fitting data, the prediction results have a large deviation from the actual value. (b) As the window slides the wavelet regressor becomes stable and the predicted value converges to the actual value.

Experimental Results

Inverse Kinematics: LMA v.s. KDL

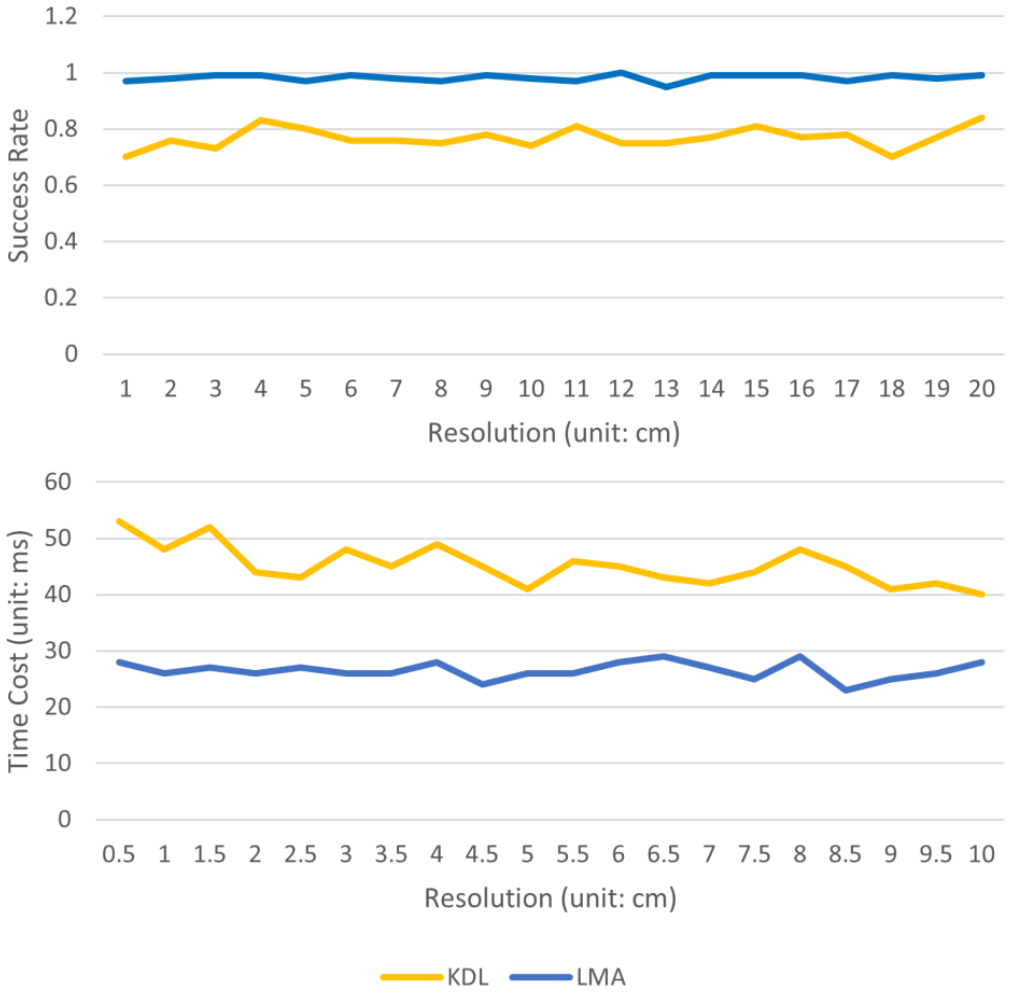


Figure: Performance comparison between KDL and LMA in accumulate success rate and average time cost of computing inverse kinematics for all of the test positions.

Experimental Results

Grasping Demonstration

Simscape Simulation

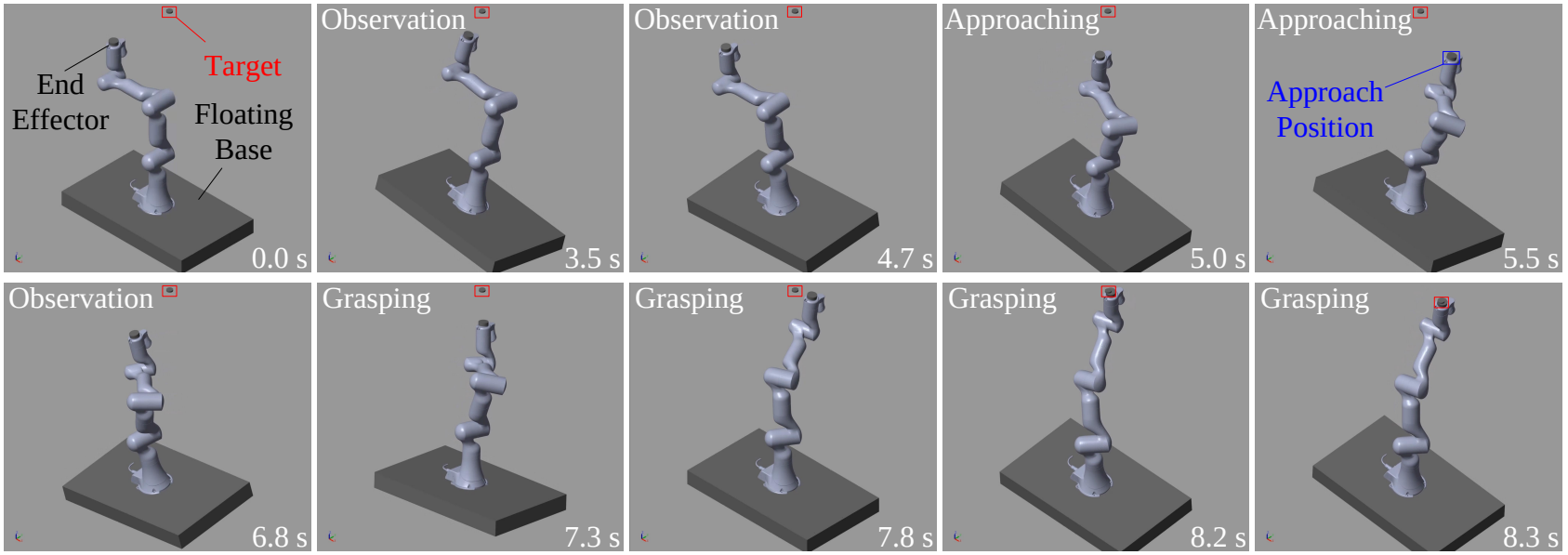


Figure: Sequential screenshot of grasping demonstration in MATLAB Simscape simulation environment.

Experimental Results

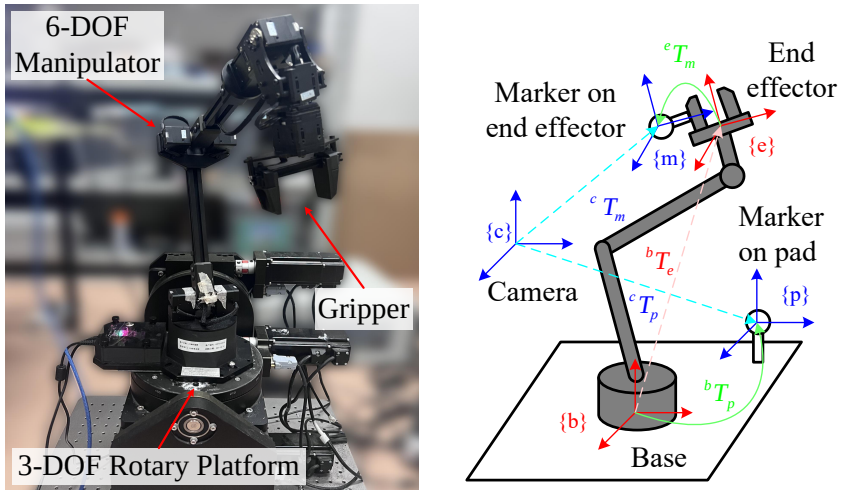
Grasping Demonstration

Physical Experiment

1. Interbotix WidowX-250s 6-DOF Robot Arm mounted on a 3-DOF rotary platform
2. OptiTrack motion capture system for localization
3. Algorithm deployed with ROS Noetic

Reason of Failure

The payload of each joint of the manipulator cannot bear the inertial force generated when shaking



(a) (b)

Figure: Experimental setup for physical demonstration. (a) A 6-DOF manipulator mounted on a 3-DOF rotary platform. (b) Illustration of eye-hand calibration for physical experiment.

Thanks for your attention!